

Utilising Numerical Weather Forecast for Planning Electricity Production in Cogeneration Plant.

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Abstract—Production of heat and electricity in the cogeneration plant depends on weather, thus forecasting production is dependent on weather forecasts. Here we present the models of the heat production based on two weather forecast models, COAMPS and UM. The linear models that are based on the predicted air temperature can explain up to 90% of variability of production and deteriorate slowly with the range of forecast. The models of heat productions that are based on UM weather forecasts significantly outperforms those that are based on the models based on the COAMPS weather forecasts. The machine learning algorithm random forest is used to improve the basic models. To this end the residuals from the linear models are predicted using various meteorological variables along with variables governing activity of city inhabitants, such as hour of the day or day of the week. This machine learning approach leads to small but significant improvement in comparison to the original model.

Index Terms—machine learning, weather forecast, heat electricity cogeneration.

I. INTRODUCTION

THE sustainable energy production requires large scale shift towards production from sources that heavily depends on the weather conditions, namely temperature, sun and thermal radiation, finally on wind. Therefore precise long-range weather forecasting becomes indispensable for planning the load balancing of energy producers. There is a substantial body of literature that concentrates on predicting wind for wind-based energy production or sunshine for solar based production, see for example [1], [2], [3], [4]. However, the utility of weather forecasts is not limited to the wind and solar power generation and it can be used by more traditional, fossil fuel based power plants. In particular this is the case for the combined heat and power (CHP) generating plants, especially if the heat production is a primary goal of the plant and electricity production is driven by the heating demands.

In Poland electricity producers are required to issue production forecast 48 hours in advance; this forecast can be later amended with 24 hour time frame. The deviations from the forecasted production are penalised economically — when utility fails to meet its forecast it has to buy the energy on the spot market, often for a much higher price than its own selling price. Correspondingly, the excessive production must be sold for a much lower price. Therefore the precise forecasting of the production is of utmost importance for producers selling energy from variable sources.

In the current study we show an application of two weather forecast models' predictions enhanced by a machine learning

model for forecasting of the heat production in the cogeneration plant.

II. DATA AND METHODS

A. Weather forecasts

ICM runs two numeric weather forecast models: COAMPS [5], [6] from the US Naval Research Laboratory and UM [7] from the UK Met Office in the semi-operational regime. Simulations with COAMPS model span 84 hours, with a grid resolution of 13km, whereas the simulations with UM model are carried on the 4km grid for 48 hours. The data for starting the model at hour 0 arrives roughly with 6 hours delay, also computations take roughly one hour, therefore the useful span of the UM model is actually closer to 40 hours for UM and 76 hours in case of COAMPS. The utility is required to deliver the production forecast 40 hours before the start of the 24 hour forecast period, while the update to the forecast should be delivered 16 hours before the forecast period. Thus only COAMPS model predictions can be used for the first forecast, whereas both COAMPS and UM models can be utilised for obtaining the forecast update. Current version of the UM model with 4km grid has been used in the operational regime from November/December 2008, whereas the COAMPS model is utilised operationally from June 2008. The results of the weather forecast simulations are archived, with most variables stored with a hourly resolution.

In the semi-operational regime, weather simulations are carried four times a day with UM model and two times a day with COAMPS model. As a result, there are numerous forecasts from the same model corresponding to the same wall-clock time; forecasts differ by the start time of the simulation. So, each variable corresponding to the atmospheric state in given time point has seven (COAMPS) or eight (UM) possible estimates with varying maturity of the forecast. They are labelled as {0h, 12h, 24h, 36h, 48h, 60h, 72h}, and {0h, 6h, 12h, 18h, 24h, 30h, 36h, 42h}, for COAMPS and UM, respectively.

B. Heat production

Hourly averages of the heat production in the cogeneration plant used for training our model was provided by our industrial partner. The heat production during the year varies considerably; in the summertime demand is more than ten times lower than during peak demand in wintertime, since there is no need for heating — thus, and because of the limited availability of the historical weather forecasts, we limited our modelling to the heating season of the year 2009. There is some variance

associated with the time when consumers start and finish their heating season, what makes is rather unpredictable; to avoid the influence of this unpredictable individual decisions data from transition periods between heating season and summer were not included. Therefore the heating season assumed in our analysis consisted of 13 weeks at the beginning of the year, starting on January 1st, and 11 weeks at the end of the year, starting at October 15.

C. Model

There are two main streams of heat loss — the conductive loss through the walls, roofs and windows, and as the sensible heat lost by ventilation. Both streams are linear functions of the temperature difference between inside and outside of the buildings. Additional effects are caused by the radiation streams — even in the wintertime in the sunny day the heat from the solar radiation absorbed by buildings may be a significant contribution to the heat balance and thus influence heat production. On the other hand, strong wind may lead to increased heat exchange by increased ventilation and better mixing of air in the vicinity of buildings.

Besides purely physical factors, there are other potentially important effects that may influence the heat production, as for instance the behaviour of the human occupants of the buildings. For example, one can expect the difference between normal business days and holidays. Another possible influence can come from a different ventilation patterns when people stay at home during the day. Finally, there are also a huge amount of subtle factors; some of them independent of the weather condition, the others depend on it in a complicated, non-linear fashion.

Therefore, the modelling in our analysis is carried out in two stages. First a simple linear model based only on the air and ground temperature predictions from a weather forecast models is constructed. Then the machine learning algorithm is used to develop a model of deviations from this linear physics-based model. To this end, a regression variant of the random forest algorithm [8] was implemented for this task. An extended set of variables were used for this correction stage: air temperature, surface temperature, wind speed, net radiative flux, hour of the day and holiday presence in that day (we counted weekends and bank holidays).

Heat production models were developed both for UM and COAMPS weather forecast models, with a set of variables adapted to the specific model output. The required electricity production forecasts cover entire range of weather forecast, therefore heat production models based on the entire available weather forecasts are required. To this end we developed models for using all seven available COAMPS forecasts and corresponding UM forecasts, if available.

To assess their quality, all considered models were produced and tested within cross-validation loop; fifteen-fold random cross-validation was used. In this protocol data is randomly split in three equal parts, two parts are used as the training set and one is used as a test set for evaluation of the error of the model. Then, the results from fifteen runs are averaged and used as the estimate of model accuracy.

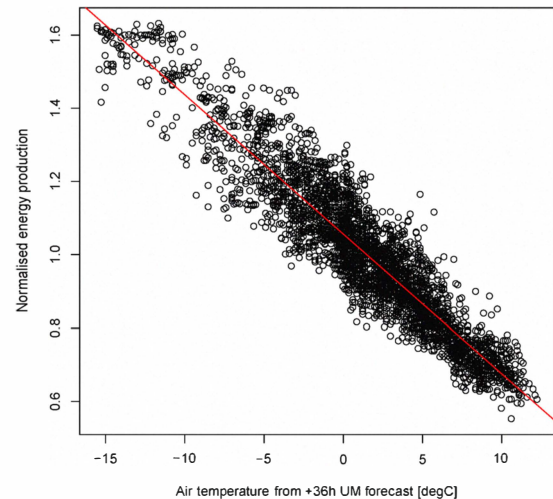


Fig. 1. Heat production versus temperature predicted by UM model

All calculations were performed in R environment for statistical computing [9]. The randomForest module [10] from R was used to build random forest models.

D. Random Forest algorithm

The random forest is a machine learning algorithm proposed by Leo Breiman; it is an ensemble method grouping CART models.

Each tree in the ensemble is built on different bagging subsample (bag) of objects and each split of the tree is constructed as a best split obtained on a randomly selected subset of descriptive variables. All individual trees are weak classifiers, however they are also only weakly co-correlated within the ensemble. In a regression task, the predictions of each CART are averaged to produce the final prediction. During the training of an RF model, an out of bag (OOB) error approximation is computed by measuring error rate for each object only using those trees for which given object was not present in their training sets (bags). Random forest also provides an estimation of variable importance computed from the estimated accuracy losses caused by nullifying information contained in a particular attribute.

III. RESULTS

As expected the heat production is strongly correlated with the ambient temperature, which is easily seen on Fig. 1. Therefore the linear model depending solely on the predicted temperature is a reasonably good forecast for heat production, see Table I.

It is rather obvious that the UM model returns significantly better forecasts than COAMPS. The quality of the forecasts deteriorates with the maturity of the forecast, nevertheless the effect is not very strong, the RMSE increases only about 10% between the most recent and the oldest forecast, both for UM and COAMPS models. The forecast based on both models has

TABLE I

RESULTS OF THE LINEAR MODEL BASED ON THE TEMPERATURE PREDICTED BY COAMPS AND UM MODELS. THE RMSE IS SHOWN AS A FRACTION OF THE AVERAGE HEAT PRODUCTION OVER ENTIRE HEATING SEASON. THE RSQ IS THE FRACTION VARIANCE EXPLAINED BY THE MODEL.

maturity	COAMPS	UM	UM + COAMPS
RMSE			
0h	0.096	0.066	0.063
12h	0.096	0.064	0.063
24h	0.093	0.070	0.066
36h	0.097	0.072	0.068
48h	0.101	-	-
60h	0.103	-	-
72h	0.104	-	-
RSQ			
0h	0.778	0.903	0.909
12h	0.781	0.906	0.912
24h	0.803	0.885	0.895
36h	0.771	0.878	0.895
48h	0.766	-	-
60h	0.752	-	-
72h	0.760	-	-

slightly lower RMSE than forecast for the UM model alone, but this effect is not significant.

The residuals after subtracting the linear model don't show visible relation to the descriptive variables, with an exception of the hour of the day variable, where slight non-linear relation is visible, see Fig. 2.

Nevertheless, the random forest algorithm is able to improve upon the linear model. The results of the random forest model compared with the best linear model are shown in the Table II.

TABLE II

RESULTS OF THE RANDOM FOREST MODEL COMPARED TO BEST LINEAR MODELS. THE RMSE IS SHOWN AS A FRACTION OF THE AVERAGE HEAT PRODUCTION OVER ENTIRE HEATING SEASON. THE RSQ IS THE FRACTION VARIANCE EXPLAINED BY THE MODEL.

maturity	COAMPS	UM	UM + COAMPS	best linear
RMSE				
0h	0.075	0.051	0.052	0.063
12h	0.082	0.053	0.055	0.063
24h	0.085	0.059	0.059	0.066
36h	0.089	0.062	0.061	0.069
48h	0.095	-	-	0.101
60h	0.097	-	-	0.103
72h	0.096	-	-	0.104
RSQ				
0h	0.865	0.942	0.938	0.909
12h	0.842	0.934	0.932	0.912
24h	0.836	0.918	0.916	0.895
36h	0.807	0.911	0.916	0.895
48h	0.791	-	-	0.766
60h	0.777	-	-	0.752
72h	0.792	-	-	0.760

TABLE III
FULL MODEL R^2

The improvement is not large, but consistent. The fraction of variance explained by the model increases by 2% or even 3%

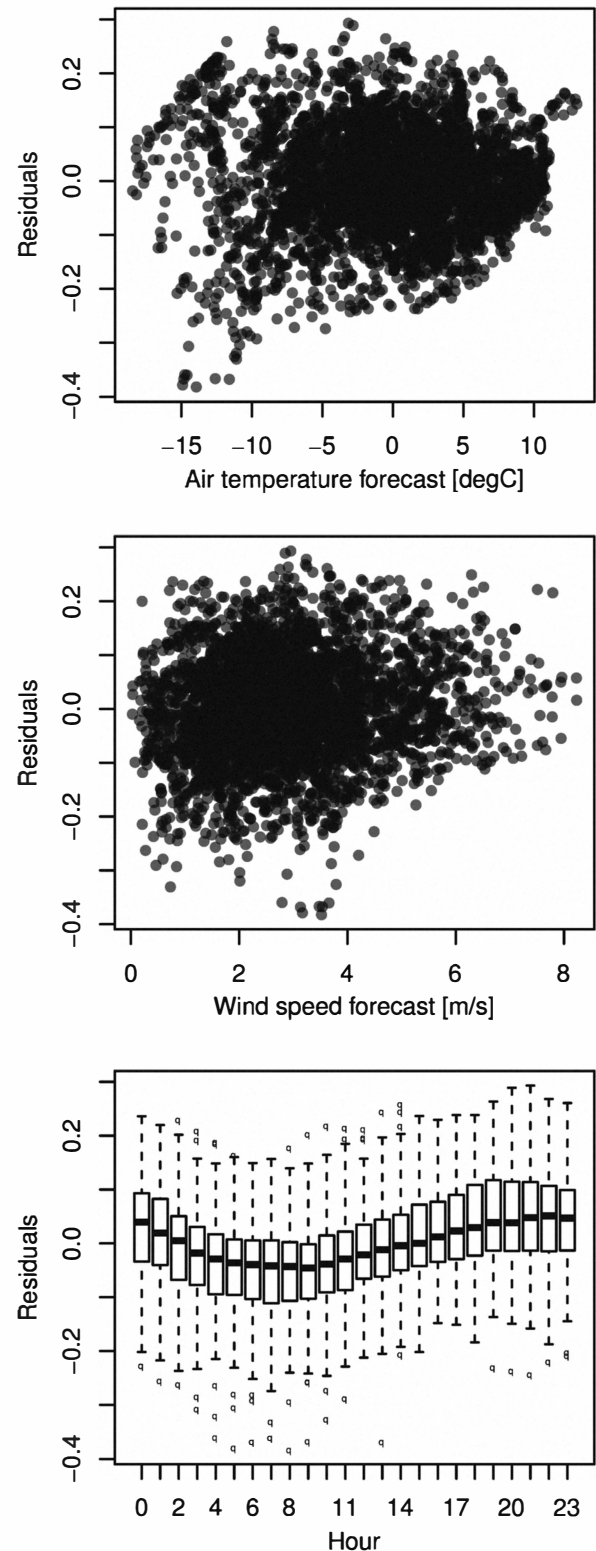


Fig. 2. Residuals from the linear model as a function of the forecast air temperature (top), wind speed (middle) and hour of the day (boxplot at the bottom).

for both models, what corresponds to the decreased RMSE is 0.6%-0.7% of the average value. These correction are a small fraction of the average production, nevertheless they represent almost 10% reduction of the modelling error in comparison

with baseline linear model.

IV. CONCLUSION

Modelling of the weather is important task for the electricity production in the cogeneration plants. The simple linear model based on the weather forecast can account for up to 91% of variability of the data, with error about 6% of the average production. However, a consistent correction of this result can be obtained with a help of a machine learning model trained on linear model residuals.

The quality of the model depends mostly on the quality of weather forecast — UM model proved to be more reliable forecasting system than COAMPS, nevertheless the relatively short span of the forecast makes it unsuitable as a base for energy production forecasts, nevertheless it can be used for issuing the corrective forecasts.

The improvement due to the machine learning correction of the physical model are smaller than the differences between models, nevertheless it is useful for including non-linear effects, for example those corresponding to the human behaviour.

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